Momentum and Its Conservation (Supplementary Problems)

1. A compact car, with mass 725 kg, is moving at 115 km/h toward the east. Sketch the moving car.
a. Find the magnitude and direction of its momentum. Draw an arrow on your sketch showing the momentum.
b. A second car, with a mass of 2175 kg, has the same momentum. What is its velocity?

2. The driver of the compact car in the previous problem suddenly applies the brakes hard for 2.0 s. As a result, an average force of 5.0×10^3 N is exerted on the car to slow it down.
a. What is the change in momentum; that is, the magnitude and direction of the impulse, on the car?
b. Complete the "before" and "after" sketches, and determine the momentum and the velocity of the car now.

3. A 7.0-kg bowling ball is rolling down the alley with a velocity of 2.0 m/s. For each impulse, shown in Figures 9-3a and 9-3b, find the resulting speed and direction of motion of the bowling ball.

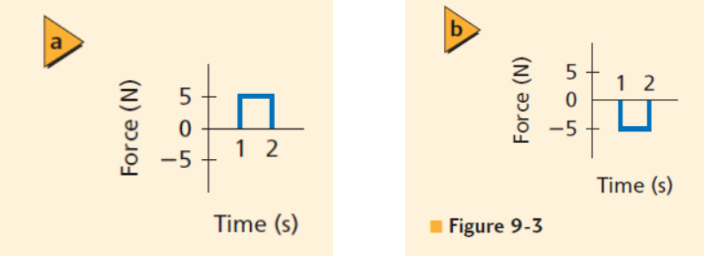


Figure 9-3

4. The driver accelerates a 240.0-kg snowmobile, which results in a force being exerted that speeds up the snowmobile from 6.00 m/s to 28.0 m/s over a time interval of 60.0 s.

a. Sketch the event, showing the initial and final situations.

b. What is the snowmobile's change in momentum? What is the impulse on the snowmobile?

c. What is the magnitude of the average force that is exerted on the snowmobile?

5. Suppose a 60.0-kg person was in the vehicle that hit the concrete wall in Example Problem 1. The velocity of the person equals that of the car both before and after the crash, and the velocity changes in 0.20 s. Sketch the problem.

a. What is the average force exerted on the person?

b. Some people think that they can stop their bodies from lurching forward in a vehicle that is suddenly braking by putting their hands on the dashboard. Find the mass of an object that has a weight equal to the force you just calculated. Could you lift such a mass? Are you strong enough to stop your body with your arms?

6. Two freight cars, each with a mass of 3.0×10^5 kg, collide and stick together. One was initially moving at 2.2 m/s, and the other was at rest. What is their final speed?

7. A 0.105-kg hockey puck moving at 24 m/s is caught and held by a 75-kg goalie at rest. With what speed does the goalie slide on the ice?

8. A 35.0-g bullet strikes a 5.0-kg stationary piece of lumber and embeds itself in the wood. The piece of lumber and bullet fly off together at 8.6 m/s. What was the original speed of the bullet?

9. A 35.0-g bullet moving at 475 m/s strikes a 2.5-kg bag of flour that is on ice, at rest. The bullet passes through the bag, as shown in Figure 9-7, and exits it at 275 m/s. How fast is the bag moving when the bullet exits?



Figure 9-7

10. The bullet in the previous problem strikes a 2.5-kg steel ball that is at rest. The bullet bounces backward after its collision at a speed of 5.0 m/s. How fast is the ball moving when the bullet bounces backward?

11. A 0.50-kg ball that is traveling at 6.0 m/s collides head-on with a 1.00-kg ball moving in the opposite direction at a speed of 12.0 m/s. The 0.50-kg ball bounces backward at 14 m/s after the collision. Find the speed of the second ball after the collision.

12. A 4.00-kg model rocket is launched, expelling 50.0 g of burned fuel from its exhaust at a speed of 625 m/s. What is the velocity of the rocket after the fuel has burned? Hint: Ignore the external forces of gravity and air resistance.

13. A thread holds a 1.5-kg cart and a 4.5-kg cart together. After the thread is burned, a compressed spring pushes the carts apart, giving the 1.5-kg cart a speed of 27 cm/s to the left. What is the velocity of the 4.5-kg cart?

14. Carmen and Judi dock a canoe. 80.0-kg Carmen moves forward at 4.0 m/s as she leaves the canoe. At what speed and in what direction do the canoe and Judi move if their combined mass is 115 kg?

15. A 925-kg car moving north at 20.1 m/s collides with a 1865-kg car moving west at 13.4 m/s. The two cars are stuck together. In what direction and at what speed do they move after the collision?

16. A 1383-kg car moving south at 11.2 m/s is struck by a 1732-kg car moving east at 31.3 m/s. The cars are stuck together. How fast and in what direction do they move immediately after the collision?

17. A stationary billiard ball, with a mass of 0.17 kg, is struck by an identical ball moving at 4.0 m/s. After the collision, the second ball moves 60.0° to the left of its original direction. The stationary ball moves 30.0° to the right of the moving ball's original direction. What is the velocity of each ball after the collision?

18. A 1345-kg car moving east at 15.7 m/s is struck by a 1923-kg car moving north. They are stuck together and move with an initial velocity of 14.5 m/s at $\theta = 63.5°$. Was the north-moving car exceeding the 20.1 m/s speed limit?

19. Golf Rocío strikes a 0.058-kg golf ball with a force of 272 N and gives it a velocity of 62.0 m/s. How long was Rocío's club in contact with the ball?

20. A 0.145-kg baseball is pitched at 42 m/s. The batter hits it horizontally to the pitcher at 58 m/s.
a. Find the change in momentum of the ball.
b. If the ball and bat are in contact for 4.6×10^{-4} s, what is the average force during contact?

21. Bowling A force of 186 N acts on a 7.3-kg bowling ball for 0.40 s. What is the bowling ball's change in momentum? What is its change in velocity?

22. A 5500-kg freight truck accelerates from 4.2 m/s to 7.8 m/s in 15.0 s by the application of a constant force. a. What change in momentum occurs? b. How large of a force is exerted?

23. In a ballistics test at the police department, Officer Rios fires a 6.0-g bullet at 350 m/s into a container that stops it in 1.8 ms. What is the average force that stops the bullet?

24. Volleyball A 0.24-kg volleyball approaches Tina with a velocity of 3.8 m/s. Tina bumps the ball, giving it a speed of 2.4 m/s but in the opposite direction. What average force did she apply if the interaction time between her hands and the ball was 0.025 s?

25. Hockey A hockey player makes a slap shot, exerting a constant force of 30.0 N on the hockey puck for 0.16 s. What is the magnitude of the impulse given to the puck?

26. Skateboarding Your brother's mass is 35.6 kg, and he has a 1.3-kg skateboard. What is the combined momentum of your brother and his skateboard if they are moving at 9.50 m/s?

27. A hockey puck has a mass of 0.115 kg and is at rest. A hockey player makes a shot, exerting a constant force of 30.0 N on the puck for 0.16 s. With what speed does it head toward the goal?

28. Before a collision, a 25-kg object was moving at +12 m/s. Find the impulse that acted on the object if, after the collision, it moved at the following velocities. a. +8.0 m/s b. -8.0 m/s

29. A 0.150-kg ball, moving in the positive direction at 12 m/s, is acted on by the impulse shown in the graph in Figure 9-16. What is the ball's speed at 4.0 s?

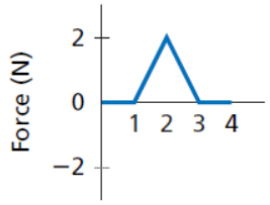


Figure 9-16

30. Baseball A 0.145-kg baseball is moving at 35 m/s when it is caught by a player.

a. Find the change in momentum of the ball.

b. If the ball is caught with the mitt held in a stationary position so that the ball stops in 0.050 s, what is the average force exerted on the ball?

c. If, instead, the mitt is moving backward so that the ball takes 0.500 s to stop, what is the average force exerted by the mitt on the ball?

31. Hockey A hockey puck has a mass of 0.115 kg and strikes the pole of the net at 37 m/s. It bounces off in the opposite direction at 25 m/s, as shown in Figure 9-17. a. What is the impulse on the puck? b. If the collision takes 5.0×10^{-4} s, what is the average force on the puck?

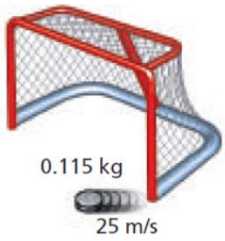


Figure 9-17

32. A nitrogen molecule with a mass of 4.7×10^{-26} kg, moving at 550 m/s, strikes the wall of a container and bounces back at the same speed.

a. What is the impulse the molecule delivers to the wall?

b. If there are 1.5×10^{23} collisions each second, what is the average force on the wall?

33. Rockets Small rockets are used to make tiny adjustments in the speeds of satellites. One such rocket has a thrust of 35 N. If it is fired to change the velocity of a 72,000-kg spacecraft by 63 cm/s, how long should it be fired?

34. An animal rescue plane flying due east at 36.0 m/s drops a bale of hay from an altitude of 60.0 m, as shown in Figure 9-18. If the bale of hay weighs 175 N, what is the momentum of the bale the moment before it strikes the ground? Give both magnitude and direction.

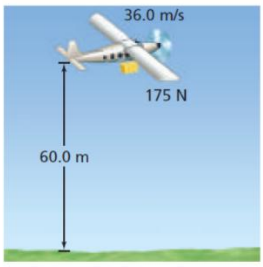


Figure 9-18

35. Accident A car moving at 10.0 m/s crashes into a barrier and stops in 0.050 s. There is a 20.0-kg child in the car. Assume that the child's velocity is changed by the same amount as that of the car, and in the same time period.

a. What is the impulse needed to stop the child?

b. What is the average force on the child?

c. What is the approximate mass of an object whose weight equals the force in part b?

d. Could you lift such a weight with your arm?

e. Why is it advisable to use a proper restraining seat rather than hold a child on your lap?

36. Football A 95-kg fullback, running at 8.2 m/s, collides in midair with a 128-kg defensive tackle moving in the opposite direction. Both players end up with zero speed.

a. Identify the "before" and "after" situations and draw a diagram of both.
b. What was the fullback's momentum before the collision?
c. What was the change in the fullback's momentum?
d. What was the change in the defensive tackle's momentum?
e. What was the defensive tackle's original momentum?
f. How fast was the defensive tackle moving originally?

37. Marble C, with mass 5.0 g, moves at a speed of 20.0 cm/s. It collides with a second marble, D, with mass 10.0 g, moving at 10.0 cm/s in the same direction. After the collision, marble C continues with a speed of 8.0 cm/s in the same direction.

a. Sketch the situation and identify the system. Identify the "before" and "after" situations and set up a coordinate system.

b. Calculate the marbles' momenta before the collision.

c. Calculate the momentum of marble C after the collision.

d. Calculate the momentum of marble D after the collision.

e. What is the speed of marble D after the collision?

38. Two lab carts are pushed together with a spring mechanism compressed between them. Upon release, the 5.0-kg cart repels one way with a velocity of 0.12 m/s, while the 2.0-kg cart goes in the opposite direction. What is the velocity of the 2.0-kg cart?

39. A 50.0-g projectile is launched with a horizontal velocity of 647 m/s from a 4.65-kg launcher moving in the same direction at 2.00 m/s. What is the launcher's velocity after the launch?

40. A 12.0-g rubber bullet travels at a velocity of 150 m/s, hits a stationary 8.5-kg concrete block resting on a frictionless surface, and ricochets in the opposite direction with a velocity of -1.0×10^2 m/s, as shown in Figure 9-19. How fast will the concrete block be moving?

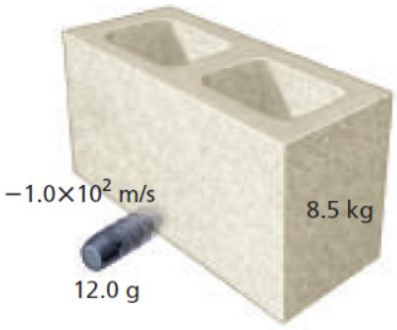


Figure 9-19

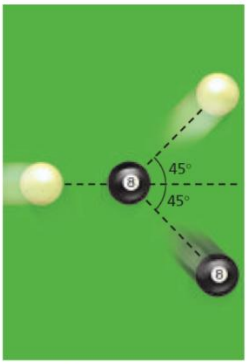


Figure 9-20

41. Skateboarding Kofi, with mass 42.00 kg, is riding a skateboard with a mass of 2.00 kg and traveling at 1.20 m/s. Kofi jumps off and the skateboard stops dead in its tracks. In what direction and with what velocity did he jump?

42. Billiards A cue ball, with mass 0.16 kg, rolling at 4.0 m/s, hits a stationary eight ball of similar mass. If the cue ball travels 45° above its original path and the eight ball travels 45° below the horizontal, as shown in Figure 9-20, what is the velocity of each ball after the collision?

43. A 2575-kg van runs into the back of an 825-kg compact car at rest. They move off together at 8.5 m/s. Assuming that the friction with the road is negligible, calculate the initial speed of the van.

44. In-line Skating Diego and Keshia are on in-line skates and stand face-to-face, then push each other away with their hands. Diego has a mass of 90.0 kg and Keshia has a mass of 60.0 kg.
a. Sketch the event, identifying the "before" and "after" situations, and set up a coordinate axis.
b. Find the ratio of the skaters' velocities just after their hands lose contact.
c. Which skater has the greater speed?
d. Which skater pushed harder?

45. A 0.200-kg plastic ball moves with a velocity of 0.30 m/s. It collides with a second plastic ball of mass 0.100 kg, which is moving along the same line at a speed of 0.10 m/s. After the collision, both balls continue moving in the same, original direction. The speed of the 0.100-kg ball is 0.26 m/s. What is the new velocity of the 0.200-kg ball?

46. A constant force of 6.00 N acts on a 3.00-kg object for 10.0 s. What are the changes in the object's momentum and velocity?

47. The velocity of a 625-kg car is changed from 10.0 m/s to 44.0 m/s in 68.0 s by an external, constant force.

a. What is the resulting change in momentum of the car?

b. What is the magnitude of the force?

48. Dragster An 845-kg dragster accelerates on a race track from rest to 100.0 km/h in 0.90 s.

a. What is the change in momentum of the dragster?

b. What is the average force exerted on the dragster?

c. What exerts that force?

49. Ice Hockey A 0.115-kg hockey puck, moving at 35.0 m/s, strikes a 0.365-kg jacket that is thrown onto the ice by a fan of a certain hockey team. The puck and jacket slide off together. Find their velocity.

50. A 50.0-kg woman, riding on a 10.0-kg cart, is moving east at 5.0 m/s. The woman jumps off the front of the cart and lands on the ground at 7.0 m/s eastward, relative to the ground.
a. Sketch the "before" and "after" situations and assign a coordinate axis to them.
b. Find the cart's velocity after the woman jumps off.

51. A 60.0-kg male dancer leaps 0.32 m high. a. With what momentum does he reach the ground? b. What impulse is needed to stop the dancer? c. As the dancer lands, his knees bend, lengthening the stopping time to 0.050 s. Find the average force exerted on the dancer's body. d. Compare the stopping force with his weight.

52. When a star that is much larger than the Sun nears the end of its lifetime, it begins to collapse, but continues to rotate. Which of the following describes the conditions of the collapsing star's moment of inertia (I), angular momentum (L), and angular velocity (ω)?

a) I increases, L stays constant, ω decreases
b) I decreases, L stays constant, ω increases
c) I increases, L increases, ω increases
d) I increases, L increases, ω stays constant

53. A 40.0-kg ice-skater glides with a speed of 2.0 m/s toward a 10.0-kg sled at rest on the ice. The ice-skater reaches the sled and holds onto it. What is the speed of the ice-skater and the sled after they collide?

a) 0.4 m/s b) 0.8 m/s c) 1.6 m/s d) 3.2 m/s

54. A bicyclist applies the brakes and slows the motion of the wheels. The angular momentum of each wheel then decreases from 3.5 kg·m²/s to 3.0 kg·m²/s over a period of 5.0 s. What is the angular impulse on each wheel?

a) -0.7 kg·m²/s b) -1.4 kg·m²/s c) -2.1 kg·m²/s d) -3.5 kg·m²/s

55. A 45.0-kg ice-skater stands at rest on the ice. A friend tosses the skater a 5.0-kg ball. The skater and the ball then move backwards across the ice with a speed of 0.50 m/s. What was the speed of the ball at the moment just before the skater caught it?

a) 2.5 m/s b) 3.0 m/s c) 4.0 m/s d) 5.0 m/s

56. When the large gear in the diagram rotates, it turns the small gear in the opposite direction at the same linear speed. The larger gear has twice the radius and four times the mass of the smaller gear. What is the angular momentum of the larger gear as a function of the angular momentum of the smaller gear? Hint: The moment of inertia for a disk is $I=1/2mr^2$, where m is mass and r is the radius of the disk.

a) $-2L_{small}$ b) $-4L_{small}$ c) $-8L_{small}$ d) $-16L_{small}$

57. A force of 16 N exerted against a rock with an impulse of 6.8 kg·m/s causes the rock to fly off the ground with a speed of 4.0 m/s. What is the mass of the rock?

a) 0.2 kg b) 0.8 kg c) 1.6 kg d) 4.0 kg

58. A 2500 kg car traveling to the north is slowed down uniformly from an initial velocity of 20.0 m/s by a 6250 N braking force acting opposite the car's motion. Use the impulse-momentum theorem to answer the following questions:

a. What is the car's velocity after 2.50 s?

b. How far does the car move during 2.50 s?

c. How long does it take the car to come to a complete stop?

59. A 0.42 kg soccer ball is moving downfield with a velocity of 12 m/s. A player kicks the ball so that it has a final velocity of 18 m/s downfield.

a. What is the change in the ball's momentum?

b. Find the constant force exerted by the player's foot on the ball if the two are in contact for 0.020 s.

60. A marble is rolling at a velocity of 1.5 m/s with a momentum of 0.10 kg.m/s. What is its mass?

61. A 0.2-kg softball is pitched horizontally at 20.0 m/s. The ball moves in the opposite direction at 25.0 m/s after it is hit by the bat during 5s.

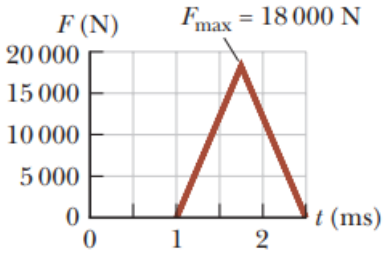
a. What is the change in momentum of the ball? What is the impulse delivered by the bat?

b. What is the magnitude of the average force that is exerted on the ball?

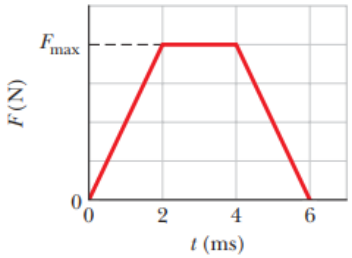
62. A force versus time graph for a baseball struck by a bat is shown in Figure below. From this curve, determine:

a. The magnitude of the impulse delivered to the ball?

b. The average force exerted on the ball?



63. The figure shows a plot of force magnitude F versus time t during the collision of a 60g Superball with a wall. The initial velocity of the ball is 25 m/s perpendicular to the wall; the ball rebounds directly back with the same speed, also perpendicular to the wall. What is F_{max}, the maximum magnitude of the force on the ball from the wall during the collision?



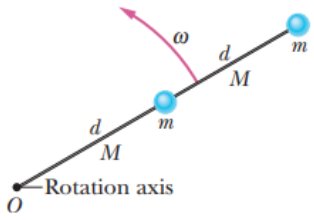
64. A ball with a moment of inertia of 0.06 kgm^2 is rolling with an angular velocity of 2 rad/s. Calculate its angular momentum.

65. what is the angular momentum of a 2.8 kg uniform cylindrical grinding wheel of radius 0.18 m when rotating at 1500 rpm?

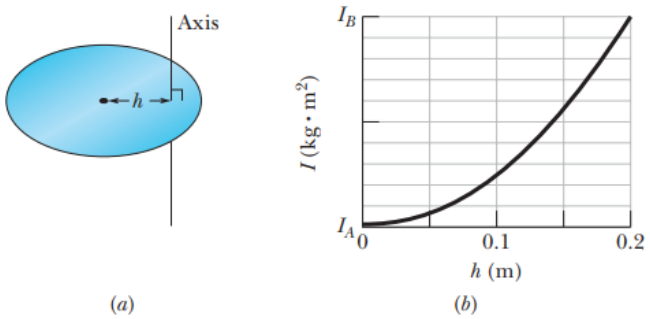
66. In Figure below, two particles of mass m = 0.85kg are fastened to each other, and to a rotation axis at O, by two thin rods, each with length d = 0.056 m and mass M = 1.2 kg. The rigid assembly rotates around point O at the angular speed ω = 0.30 rad/s. About O, what are:

a. the moment of inertia of the assembly?

b. The magnitude of the angular momentum of the assembly?

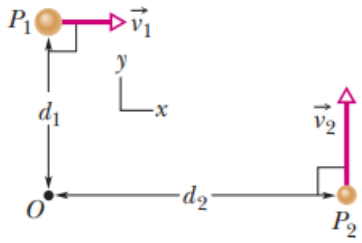


67. Figure (a) below shows a disk that can rotate about an axis at a radial distance h from the center of the disk. Figure (b) gives the moment of inertia I of the disk about the axis as a function of that distance h, form the center of out to the edge of the disk. The scale on the I axis is set by $I_B = 0.15\ kg.m^2$. (c) What is the mass of the disk? (d) Find the angular momentum of the disk if it rotates with angular speed $\omega = 10\ rad/s$?



68. A ball with a moment of inertia of 0.06 kgm^2 is rolling with an angular velocity of 2 rad/s. Calculate its angular momentum.

69. In the instant of Figure below, two particles move in an xy plane. Particle P_1 has mass 6.5 kg and speed v_1= 2.2 m/s and it is at distance d_1 = 1.5 m from point O. Particle P_2 has mass 3.1 kg and speed v_2= 3.6 m/s and it is at distance d_2 = 2.8 m from point O. What are the magnitude and direction of the net angular momentum of the two particles about O?



70. Determine the angular momentum of the Earth rotates about its rotation axis (assume the Earth is a uniform sphere). Given: The Earth has mass 6×10^{24} kg and radius 6.4×10^{6}m.

71. A 25.0 g marble sliding to the right at 20.0 cm/s overtakes and collides elastically with a 10.0 g marble moving in the same direction at 15.0 cm/s. After the collision, the 10.0 g marble moves to the right at 22.1 cm/s. Find the velocity of the 25.0 g marble after the collision.

72. A 3.00 kg mud ball has a perfectly inelastic collision with a second mud ball that is initially at rest. The composite system moves with a speed equal to one third the original speed of the 3.00 kg mud ball. What is the mass of the second mud ball?